Practical reproducibility for data analysts

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2024

Introduction

Analyses fail to reproduce mainly because the computational environment is not recorded. Pinning package versions in a lock file allows another analyst to rebuild the environment, although lock files alone do not capture operating-system libraries.

Version control

Version control records every change to analysis code, which makes it possible to identify exactly which code produced a result. However, version control does not track large data files well, so data versions must be managed separately.

Conclusion

No single practice guarantees reproducibility; practices must be combined.